

Broker AI Backend - Deployment & Configuration Guide

Overview

You have successfully built a **Hybrid AI Trading Broker** with:

-  Technical Analysis (RSI, MACD, MA50/200, Volume)
 -  Automated Signal Generation (3+ convergent indicators)
 -  Telegram Integration with Interactive Buttons
 -  Alpaca Order Execution
 -  Real-time P&L Tracking
 -  Complete Trade History Database
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Pre-Deployment Checklist

Before deploying, you need to gather these credentials:

1. Alpaca Trading API (Stock Trading)

- Sign up at <https://alpaca.markets>
- Create an API key in your dashboard
- You'll need:
 - `ALPACA_API_KEY` - Your API key
 - `ALPACA_SECRET_KEY` - Your secret key

2. Telegram Bot Token (Notifications)

- Open Telegram and chat with [@BotFather](https://t.me/botfather) (<https://t.me/botfather>)
- Send: `/newbot`
- Follow the prompts to create a bot
- You'll receive:
 - `TELEGRAM_BOT_TOKEN` - The bot token
- Find your chat ID:
- Open your bot and send any message
- Visit: `https://api.telegram.org/bot{TOKEN}/getUpdates`
- Look for `"chat": {"id": xxxx}`
- This is your `TELEGRAM_CHAT_ID`

3. API Security Key (Optional but recommended)

- Generate a secure random key: `openssl rand -hex 32`
 - Set as `ANALYSIS_API_KEY` to protect the `/analysis/run` endpoint
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Deployment Steps

Step 1: Deploy to Production

1. Click the **Deploy** button in the UI
2. Your service will be deployed to a public URL like:
`https://your-broker-ai.abacusai.app`
3. Save this URL - you'll need it for webhook registration

Step 2: Configure Environment Variables

After deployment, set the environment variables in the deployment console:

```
# Alpaca Trading
ALPACA_API_KEY=your_alpaca_api_key_here
ALPACA_SECRET_KEY=your_alpaca_secret_key_here

# Telegram Bot
TELEGRAM_BOT_TOKEN=your_bot_token_here
TELEGRAM_CHAT_ID=your_chat_id_here

# API Security
ANALYSIS_API_KEY=your_secure_api_key_here
```

Step 3: Register Telegram Webhook

Once deployed, register your Telegram webhook:

```
curl -X POST https://api.telegram.org/bot{TELEGRAM_BOT_TOKEN}/setWebhook \
-H "Content-Type: application/json" \
-d '{
  "url": "https://your-deployed-url/webhook/telegram",
  "allowed_updates": ["callback_query"]
}'
```

Replace:

- `{TELEGRAM_BOT_TOKEN}` with your actual token
- `your-deployed-url` with your production URL

Step 4: Create Analysis Cron Job

To run technical analysis every 15 minutes, create a cron job that calls:

```
POST https://your-deployed-url/analysis/run
Headers: Content-Type: application/json
Body: { "api_key": "your_ANALYSIS_API_KEY" }
```

Using Abacus AI Cron Job Tool:

```
Name: "Broker AI Technical Analysis"
Schedule: "0 */15 * * *" (every 15 minutes)
Endpoint: /analysis/run
Method: POST
Payload: { "api_key": "your_ANALYSIS_API_KEY" }
Headers: { "Content-Type": "application/json" }
```

API Endpoints

Analysis

- **POST /analysis/run** - Execute technical analysis (called by cron)
- Request: `{ "api_key": "string" }`
- Response: `{ "success": boolean, "signals_generated": number }`

Signals

- **GET /signals** - Get recent signals (query: `?limit=20`)
- **GET /signals/:symbol** - Get signals for specific symbol

Orders

- **GET /orders** - Get all orders (query: `?limit=50`)
- **GET /orders/symbol/:symbol** - Get orders for symbol
- **POST /orders/:id/authorize** - Execute pending order
- **DELETE /orders/:id** - Cancel order

Performance & P&L

- **GET /performance** - Get closed trades with P&L
- **GET /performance/symbol/:symbol** - P&L by symbol
- **GET /performance/stats/summary** - Overall statistics

Configuration

- **GET /config** - Get all symbol configurations
- **GET /config/:symbol** - Get specific configuration
- **POST /config/:symbol** - Create/update configuration
- **POST /config/:symbol/disable** - Disable monitoring

Telegram Webhook

- **POST /webhook/telegram** - Telegram updates (callback queries)
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How the System Works

1. Technical Analysis (Every 15 Minutes)

```

Cron Job Triggers /analysis/run
↓
Fetch market data from Alpaca (250 recent bars)
↓
Calculate indicators:
• RSI (14 period)
• MACD (12/26/9)
• Moving Averages (50, 200)
• Volume analysis
↓
Detect convergence (3+ indicators align)
↓
If signal generated:
• Save to database
• Send Telegram notification with buttons
  
```

2. User Authorization via Telegram

```

User receives signal notification
↓
Clicks BUY or SELL button
↓
Bot receives callback query
↓
/webhook/telegram processes it
↓
Create order in database
↓
Execute on Alpaca
↓
Send confirmation to user
↓
Start tracking P&L
  
```

3. Performance Tracking

```

BUY order executes
↓
Create performance record (OPEN status)
↓
Monitor price changes
↓
When SELL order executes:
• Calculate P&L amount
• Calculate P&L percentage
• Calculate hold duration
• Mark as CLOSED
  
```

Configuration Parameters

Each symbol has customizable parameters:

```
{
  "symbol": "AAPL",
  "enabled": true,
  "analysis_interval_min": 15,

  "rsi_period": 14,
  "rsi_overbought": 70,
  "rsi_oversold": 30,

  "macd_fast_period": 12,
  "macd_slow_period": 26,
  "macd_signal_period": 9,

  "ma50_period": 50,
  "ma200_period": 200,

  "volume_threshold_pct": 120,
  "required_convergence": 3
}
```

Update Configuration

```
curl -X POST https://your-url/config/AAPL \
-H "Content-Type: application/json" \
-d '{
  "rsi_overbought": 75,
  "required_convergence": 4
}'
```

Telegram Signal Format

When a signal is generated, you'll receive:

```
🟢 BUY Signal Generated!

📊 Symbol: AAPL
💰 Current Price: $150.25

📈 Technical Analysis:
• RSI: 28.45 (Oversold)
• MACD: -2.3450 (Below signal)
• MA50: $151.00
• MA200: $155.00

📝 Analysis:
RSI is oversold (<30); Price is below MA50

[✅ BUY] [❌ Cancel]
```

Click  **BUY** → Order executes on Alpaca

Database Schema

All data is stored in PostgreSQL:

Signals - Technical analysis results

Orders - Buy/Sell order history

Performance - P&L tracking

Configuration - Symbol settings

Access data via API endpoints or directly via Prisma queries.

Testing

Test Health Check

```
curl https://your-url/health
```

Test Configuration

```
curl https://your-url/config
```

Test Analysis (Manual Trigger)

```
curl -X POST https://your-url/analysis/run \
-H "Content-Type: application/json" \
-d '{"api_key": "your_ANALYSIS_API_KEY"}'
```

View Signals Generated

```
curl https://your-url/signals?limit=10
```

Important Notes

Market Hours

- Alpaca live trading is during market hours (9:30 AM - 4:00 PM EST)
- Paper trading is available 24/7 (use paper trading account for testing)

Rate Limiting

- Alpaca API has rate limits (~200 requests/minute)
- The cron job runs every 15 minutes (safe)
- If monitoring many symbols, adjust timing as needed

P&L Calculation

- $P\&L = (Exit\ Price - Entry\ Price) \times Quantity$

- $P\&L \% = ((Exit\ Price - Entry\ Price) / Entry\ Price) \times 100$
- Percentages are stored with 4 decimal precision

Security

- API keys are stored securely in environment variables
- Always use HTTPS in production
- Protect your `ANALYSIS_API_KEY` from unauthorized cron calls

API Documentation

Full interactive API documentation available at:

```
https://your-deployed-url/api-docs
```

All endpoints are documented with:

- Request/response examples
- Parameter descriptions
- Status codes

Troubleshooting

Telegram Webhook Not Working

1. Verify webhook URL is accessible
2. Check `TELEGRAM_BOT_TOKEN` is correct
3. Verify bot has message permissions
4. Check server logs: `GET /prod logs`

No Signals Generated

1. Ensure at least 250 price bars available
2. Check if 3+ indicators truly converge
3. Verify symbol exists on Alpaca
4. Check configuration is enabled

Order Execution Fails

1. Verify Alpaca credentials are correct
2. Check account has sufficient buying power
3. Ensure market is open (if live trading)
4. Check order quantity is valid

Performance Not Updating

1. Ensure SELL orders are created
 2. Check performance records have both entry and exit prices
 3. Run: `GET /performance/stats/summary` to verify
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Support

For issues, check:

1. API Documentation: `/api-docs`
 2. Server Logs: Via Abacus AI console
 3. Database queries: Via Prisma studio
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Next Steps

1. ☒ Deploy the service (click Deploy button)
2. ☒ Configure environment variables
3. ☒ Register Telegram webhook
4. ☒ Set up 15-minute cron job
5. ☒ Enable monitoring for first symbol
6. ☒ Monitor first signal generation
7. ☒ Test order execution via Telegram

You're ready to go live! 🚀